

Ratio/Multiplicative Cognition

Literature review

Fish and Numerosity

From [this](#) paper, we see that fishes are capable of non-symbolic numerosity cognition. They are able to distinguish larger and smaller quantity (discriminate larger shoals vs smaller ones). They show ratio-dependence especially when it comes to higher quantities which is consistent with our findings in other higher order organisms (approximate number systems). For smaller quantities these fishes seem to be less ratio dependent as opposed to larger quantities. However, it is also true that fishes use non-numerical cues such as surface area, density and other continuous variables when making decisions but when these factors are controlled for, they are still able to perform numerical tasks. This indicates that they have a true “number sense”.

Ratio abstraction in chicks

[This paper](#) shows that newly hatched domestic chicks can extract and generalize proportional relations across different absolute numerosities. The arrays consisted of green and red dots, and the chicks were rewarded for selecting the array with the 4:1 proportion. For example, the 4:1 ratio could be represented by 32 green dots and 8 red dots, while the 2:1 ratio could be represented by 24 green dots and 12 red dots. After training, the chicks were tested with novel arrays that maintained the 4:1 ratio but with different absolute numbers and spatial configurations, such as 20 green dots and 5 red dots. Importantly, chicks transferred the learned proportion to novel arrays with different absolute numbers, demonstrating abstraction of the ratio (a/b) rather than memorization of particular absolute values. The work used careful stimulus controls (varying absolute numerosities) and this experimental design (train to a proportion, test on different absolute values) is a direct template for testing multiplicative rules such as doubling.

Proportion discrimination in pigeons

[This study](#) shows that pigeons are capable of discriminating continuous ratios using coloured bars. Here, pigeons were presented with bars that were painted in 2 colours (red and green) and were trained to pick the bar with the highest proportion of the positive colour (either red or green depending on the trial). Post operant conditioning, it was found that pigeons can differentiate the ratios and consistently pick out the bar with the higher proportion of the positive colour. Similar to fish studies, the accuracy fell when the proportions became more similar (1:2 vs 6:7). The novel finding here is that they are able to distinguish smaller and larger ratios in continuous proportion displays and not only in absolute numbers (example: distinguishing 2 shapes and 4 shapes vs 4 shapes and 8 shapes).

Comparing discrete ratios in primates

Similar to the chicks, [this study](#) shows that Rhesus macaques can compare discrete ratios directly. They can treat each stimulus as a ratio (a/b) and choose between two ratios, even when both numerators and denominators vary. However, the experimental design is different. Each

stimuli has negative and positive shapes. The task is to pick the stimulus that has a greater ratio of positive to negative shapes. Between two stimuli that contain 4 negative shapes + 8 positive shapes and 1 negative + 3 positive shapes, the monkey needs to choose the latter (3:1 is a bigger ratio than 8:4/2:1). The monkeys were able to consistently pick the correct stimuli even when novel ratios were introduced during the testing trials.

The experimental paradigms used in fish, chicks, pigeons, and primates provide a comparative framework for studying ratio and multiplicative cognition in zebrafish. Fish studies show that zebrafish can discriminate shoal sizes and exhibit ratio-dependence when continuous variables are controlled, indicating a non-symbolic number sense. However, these studies primarily test relative magnitude (larger vs. smaller) rather than abstract ratio rules. Chick experiments demonstrate that training subjects on a target proportion and then testing them with novel absolute numerosities isolates ratio abstraction from rote learning. This approach can be applied to zebrafish by training them to prefer a specific visual ratio (e.g., 2:1) and then testing novel sets (e.g., 6:3, 12:6) to assess internalization of the ratio rule. Pigeon studies show that continuous perceptual ratios, such as color proportions, can be generalized, emphasizing the need to distinguish discrete numerical reasoning from reliance on perceptual cues like area or density. Similarly, rhesus macaques can compute and compare complex ratios by treating numerators and denominators relationally, rather than focusing on total quantity. Applying these methods to zebrafish allows testing of higher-order multiplicative reasoning and moves beyond simple numerosity discrimination, providing a rigorous framework to examine whether zebrafish share an abstract, ratio based cognitive system with other species.

Methods (Preliminary Protocol)

General setup

Model Organism: adult zebrafish (*Danio rerio*), N = 16–24 (balanced sexes), naïve to numerosity testing. Housed in standard conditions, food withheld for 2–4 h pre-session to motivate foraging reward.

Apparatus: rectangular test tank with two choice zones at either end; computer monitor or laminated stimulus cards behind transparent barriers; automated video tracking or experimenter-scored choices. Choice = first entry into choice zone / touching stimulus. Reward = small food pellet delivered at choice location.

Stimuli: 2D shapes (black filled circles/squares) on a neutral background. When needed, control continuous cues by randomizing dot size, total area, density, and spatial distribution across trials; equate cumulative area across options when testing discrete counts.

Inter trial interval and number of trials?

Experiment 1

Goal: train fish to choose the stimulus that yields a fixed multiplicative relation (e.g., $1 \rightarrow 2$, i.e. pick the card containing twice the number of shapes as the sample).

Training: On each trial present a sample card in the center showing a small set (denominator), e.g. “1” or “2” shapes. After 2s delay, present two choice cards simultaneously at ends: one card shows the correct multiplicative match (eg: if 2 shapes are shown initially, correct answer is 4) and the other shows a foil (such as 3 or 5). Reward fish for choosing the multiplicative match. Randomize which side contains the correct choice. Start with large numerical differences (eg: 1 vs 4), then start presenting more similar ratios (eg: 6 vs 7).

Test trials: After testing, present novel denominator magnitudes never used in training (eg: sample = 4 or sample = 8) with choice options that test the 1:2 rule (eg: for sample 4, choices 8 vs 6).

Controls: equate continuous variables (total area, item size and density) across choices; include trials in which total area or density conflicts with correct ratio to check cue reliance. Analyze % correct on transfer and control trials.

Experiment 2

Goal: train zebrafish to select the array with the higher positive:negative ratio regardless of absolute counts.

Stimuli: Each array contains two symbol types (eg: green = positive, red = negative). Ratios vary (target ratio = 2:1 vs foil = 1:2, etc.). Absolute numerosity of each array is varied across trials (e.g., 8:4 vs 6:3 vs 2:1), so the same ratio appears with different totals.

Training: Present two arrays simultaneously. Reward selection of the array with the higher positive:negative ratio. Randomize absolute totals and sides of the correct option.

Test trials: Test with novel absolute counts that preserve the target ratio (e.g., trained on 4:2 vs 3:1, test on 12:6 vs 9:3) and with novel spatial arrangements. Examine whether fish choose by ratio rather than specific numerosity.

Controls: include trials where continuous cues (positive area, total area) are dissociated from ratio (e.g., smaller number of larger positive items vs larger number of smaller positives) to assess cue dependence.

Experiment 3

Goal: determine whether zebrafish respond to continuous proportional cues (area/coverage or color proportion) rather than discrete counts.

Stimuli: Rectangular bars or patches divided into two colors (e.g., blue = positive, grey = negative). Bars vary in the proportion of positive color (e.g., 66% vs 50%), while total bar size remains constant. Alternatively, we can use dot arrays but manipulate cumulative positive areas rather than count.

Training: Two-alternative forced choice: reward fish for choosing the bar with the higher positive proportion. Start with large proportion differences (e.g., 80% vs 20%) and then slowly decrease the difference.

Probe tests: Present choice pairs where discrete number and continuous proportion conflict (e.g., more positive dots but smaller cumulative positive area) to determine which cue fish prioritize. Also test transfer to novel shapes/sizes (same proportion but different absolute area) to test abstraction of proportion.

Analysis: compare accuracy across proportion distances (Weber/rate dependence) and conflict trials to evaluate reliance on continuous vs discrete cues.

Problems

Unsure if the zebrafish will be able to do any of this. I don't think it is likely given that none of the studies I came across were done on fish.